Peoplesoft ID:

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EEE3094S 2024: Deferred Exam - Memo

- The exam is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write "see attached pages" by the question) and make sure they are marked with your peoplesoft ID.
- There are four questions totalling 103 marks, but the maximum mark is 100.
- You have 15 minutes reading time and 3 hours to complete the exam.
- The last page is a formula sheet.

Question 1 [15 marks]

The information sheet following this question shows the unit step response and frequency response of an unknown system. (Note that this sheet has plots on both sides.)

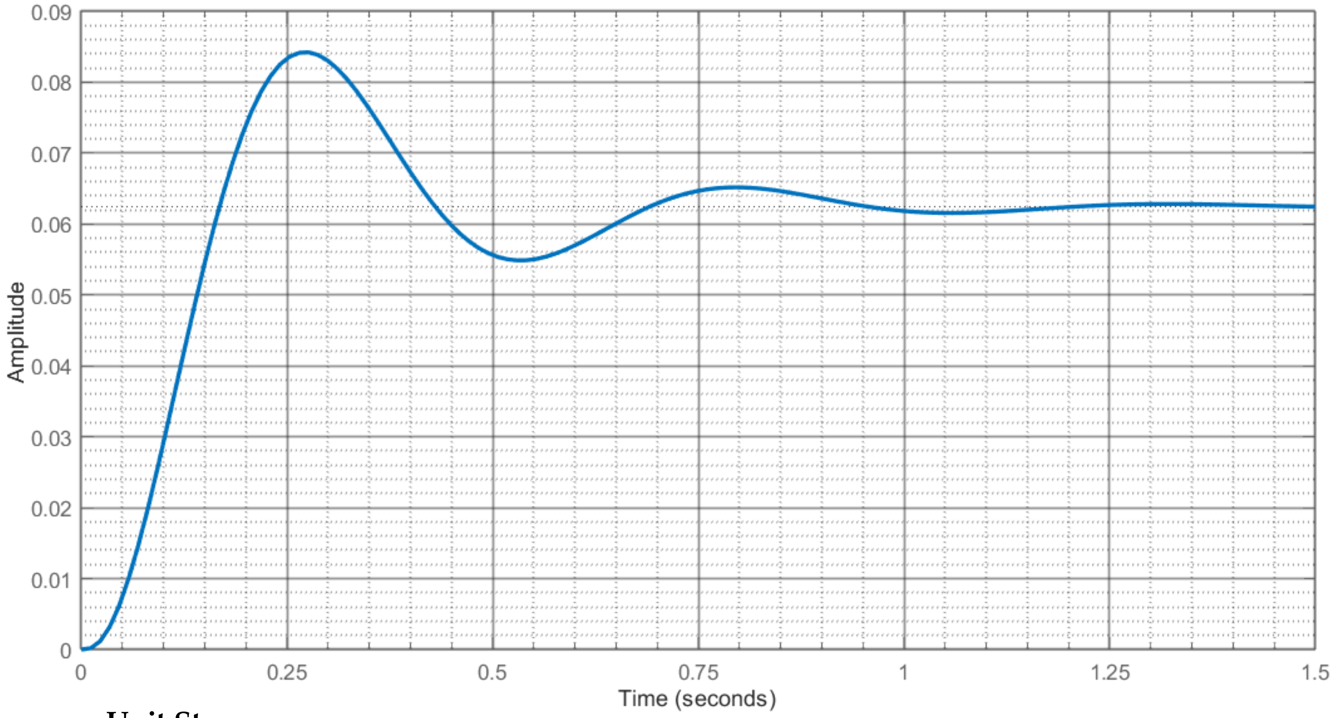
a) Explain **three** features that show that these responses come from the same system. [9 marks]

b) Suggest a second-order transfer function model for this system. [3 marks]

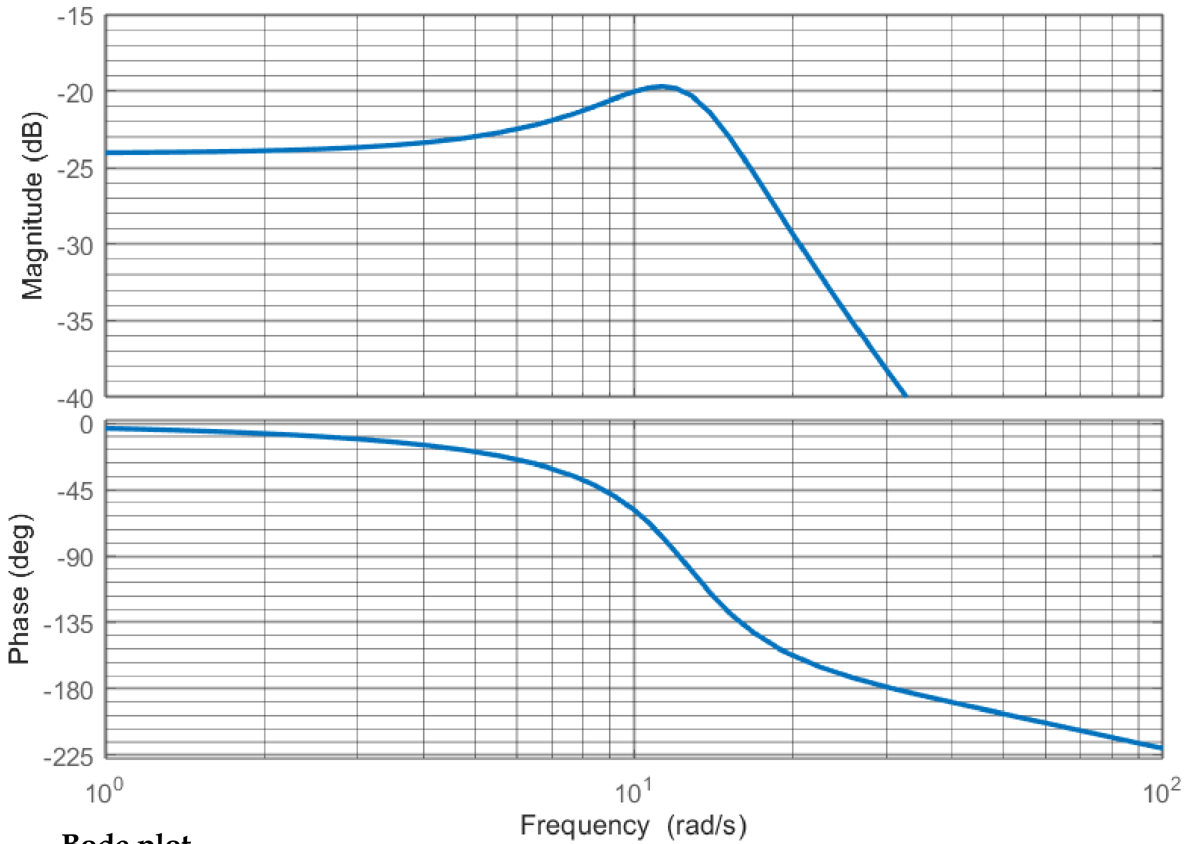
c) Is this system actually second order? If not, would you consider a second-order model to be an acceptable approximation? Explain. [3 marks]

Question 1 – Information

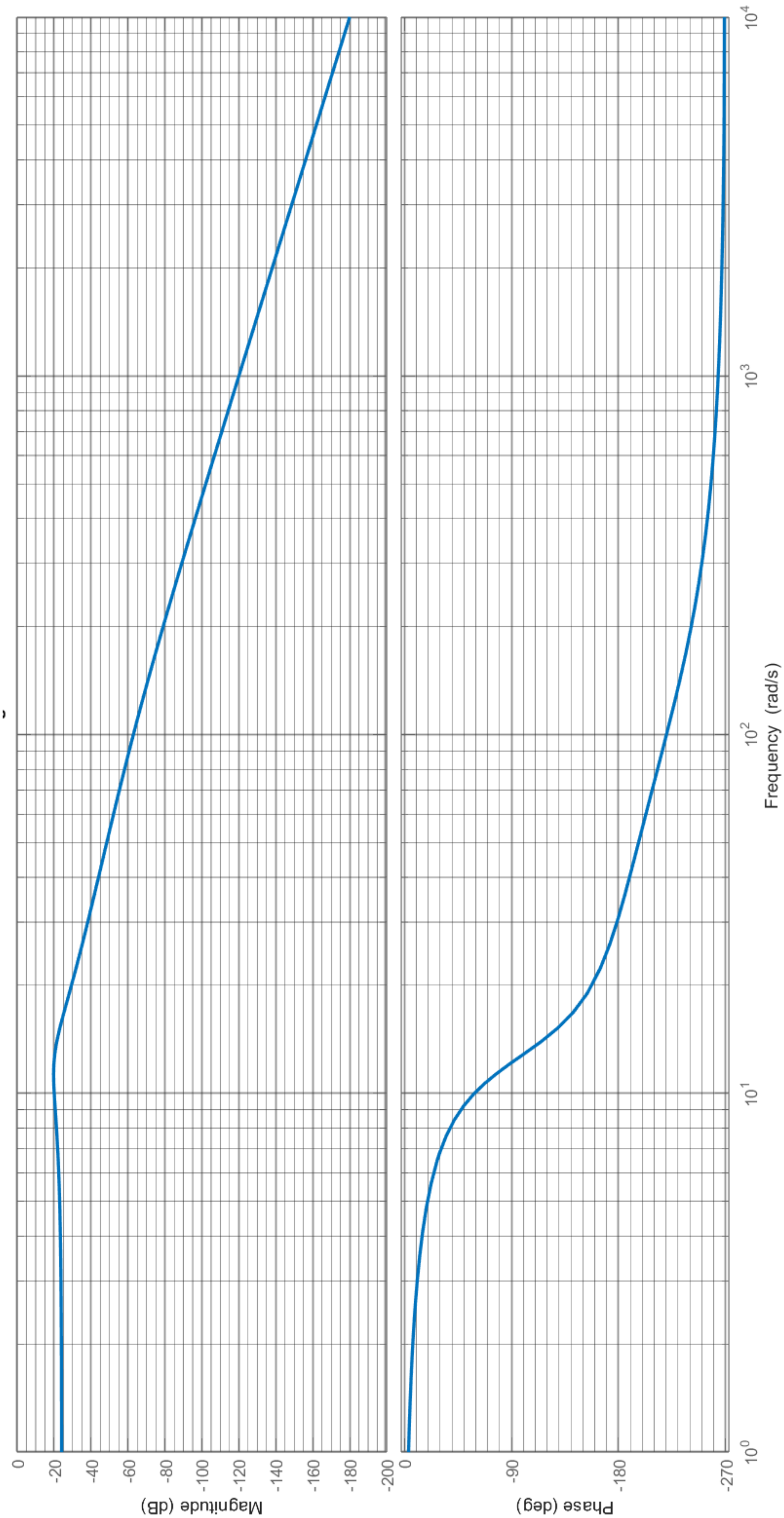
(For convenience, you may remove this sheet by tearing or cutting on the dotted line)



Unit Step response



Bode plot



Bode plot, expanded.

Solutions:
a) (1) The gain: [1 mark]
The frequency plot shows a low-frequency gain of -24 dB (0.06). This is consistent with the final value of the step response, which is around 0.06. [2 marks]

(2) The damping ratio: [1 mark]
The step response has a peak magnitude of 0.084, and final value of 0.062, giving a 35% overshoot. Using the given formula, this is produced by a damping ratio $\zeta = 0.32$.

This damping ratio should give a peak in the magnitude Bode plot with a magnitude of $\frac{1}{2\zeta} = 1.56$, or 3.9 dB more than the DC gain. The peak in the given Bode plot is around 4 dB greater than the DC value.
[2 marks]

(3) The natural frequency: [1 mark] (also: settling time, or any other indicator of the time constant)
The first peak occurs at around $t = 0.275$ seconds, giving a damped frequency $\omega_d = \frac{\pi}{0.275} = 11.42$ rad/s. This gives a natural frequency of $\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} = \frac{11.42}{\sqrt{1-0.35^2}} = 12.2$ rad/s. This corresponds to the position of the peak in the magnitude Bode plot. [2 marks]

b) The basic form of a second-order transfer function is $\frac{A\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$. Substituting in the values of gain, damping ratio and natural frequency estimated in part a gives $\frac{7.8}{s^2+8s+130.4}$. [3 marks]

c) No, we can see from the expanded Bode plot that there is an additional pole, as the roll-off of -60 dB/decade and minimum phase of -270 degrees are both consistent with a third-order system. This pole is at a much higher frequency than the second-order pole pair, however (greater than 100 rad/s, based on where the phase Bode plot intersects -225 degrees), so the slower poles will dominate the response. The second-order model is therefore an acceptable approximation of the system's behaviour. [3 marks]

Question 2 [20 marks]

Your body is a wonderland… of feedback control! Even when you aren't doing something that makes this as obvious as balancing a ball on your head, feedback processes are working throughout your body to maintain the conditions you need to live.

An example is the regulation of blood glucose by the endocrine and digestive system.

a) Using the information in section 1 of the information sheet following this question, construct a block diagram modelling the control of blood glucose in the body. (You may include blocks representing conditional functions e.g. "$\geq x$", to create unidirectional paths.) [12 marks]

b) Your friend has type 1 diabetes, meaning that the beta cells in her pancreas do not produce enough insulin. She must therefore inject herself with insulin after meals. One day, she forgets the monitor she uses to measure her glucose levels at home. "Don't worry," she tells you, "I'm having the same thing for lunch I had yesterday, so I'll just inject the same amount."

   What type of control does this represent? Is this safe? Explain. [4 marks]

c) Consider section 2 of the information sheet, concerning glycemic index. Based on this information, explain two important things can you determine about the order and properties of this system. [4 marks]

Question 2 – Information

(For convenience, you may remove this sheet by tearing or cutting on the dotted line)

Section 1 – Overview of blood glucose regulation

- Receptors $R(s)$ in the pancreas detect the current level of glucose in the bloodstream, $y(s)$, and compare it to the ideal range $r(s)$.

- If blood glucose is too high, the beta cells $\beta(s)$ in the pancreas raise the level of the hormone, insulin $i(s)$, causing the body's cells $C(s)$ to absorb glucose from the blood.

- If blood glucose is too low, the alpha cells $\alpha(s)$ in the pancreas raise the level of the hormone, glucagon $g(s)$, which triggers the liver $L(s)$ to break down stored glycogen, releasing glucose into the blood.

- Low glucose levels also cause the nervous system to instigate hunger $H(s)$, which we will assume leads to an intake of glucose from food.

- We can therefore consider the blood glucose level at any given time to be a combination of three elements: glucose absorption by the cells $a(s)$, glucose release from the breakdown of energy stores $b(s)$, and intake of glucose from food $f(s)$.

- Factors such as illness, stress, physical activity and pregnancy can affect blood glucose levels. These fluctuations can be modelled as an unknown external disturbance $d(s)$.

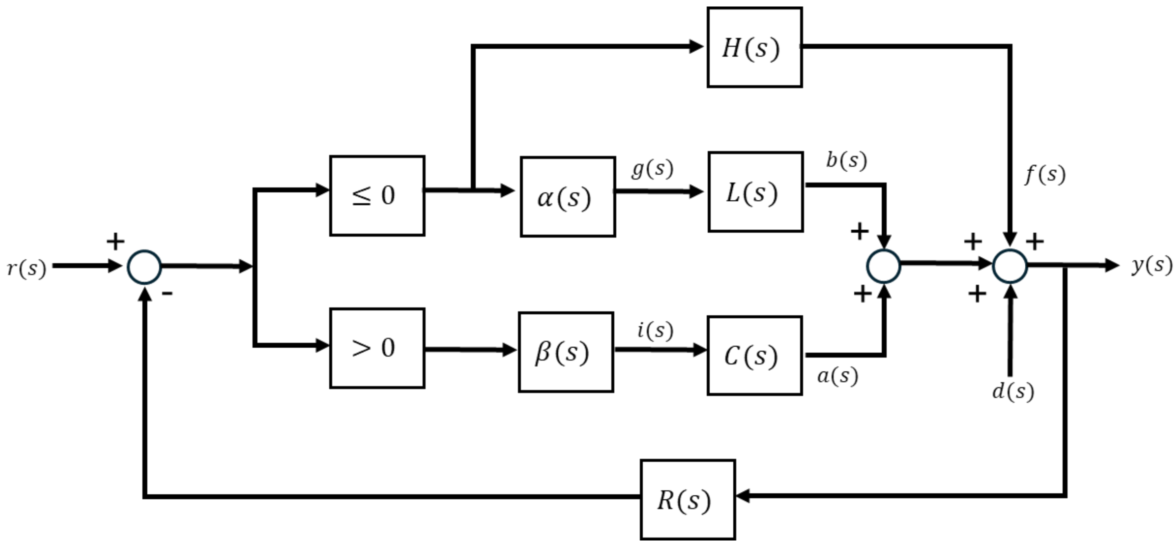
Section 2 – Glycemic Index

The glycemic index (GI) of food is a measure of how quickly the food can be broken down to release glucose. High GI foods such as sweets, refined carbohydrates and some fruits release quickly, leading to large spikes in blood glucose. Low GI foods such as whole grains and protein sources break down more slowly, leading to a smaller peak in blood glucose level sustained over a longer time.

While both high and low GI foods can provide the energy we need to survive, eating a high-GI diet puts additional stress on the body by creating large swings in blood glucose. The large spikes following the intake of these foods lead to large releases of insulin, and correspondingly, large "crashes" in blood glucose level that lead to it falling below the ideal range. These crashes then trigger cravings for more high-GI foods to raise glucose quickly, and the cycle repeats.

Solutions:

a)



marking:

feedback structure: 2

high glucose path: 2

low glucose path: 2

hunger path: 2

feedback path: 2

summing junction: 1

disturbance: 1

b) This is an example of open loop control [1 mark], as the friend does not have information about the current state of her blood glucose when applying the control action [1 mark]. This is not safe [1 mark] as disturbances may cause the state, and therefore, the response, to be different even though the input is the same.

c) The paragraph shows that this system has overshoot. [1 mark] This indicates that at least a second-order model is required for the system, as oscillation is a second-order phenomenon. [1 mark] Systems with oscillation are classified as underdamped or undamped, and have damping ratios between 0 and 1. [1 mark for any of these statements about the damping.]

Question 3 [38 marks]

Your crush has just agreed to go with you to a fun new barcade in your area! You want to win their heart by winning them a prize, and the fluffy toys literally up for grabs in the claw machine are the cutest ones on offer. The only problem: you suck at catching anything!

But no worries: rather than relying on your inadequate reflexes to control the claw, you decide to design a feedback controller to get it to where it needs to be. This will also create an opportunity to impress your date with your control engineering expertise. Surely, nobody could resist!

You go to the barcade to conduct an investigation into the machine to help you in your design. The results of your investigation are given on the information sheet following this question.

a) Use the provided information to derive the following model relating the vertical position of the claw y to the input current from the user inputs, i:

$$G = \frac{16.28}{s(s + 4.17)}$$

[12 marks]

b) If you use proportional control, what is the shortest possible time it will take for the closed-loop system to move the claw to within 2% of its final position? [4 marks]

c) The heaviest toy you could pick up weighs around 0.3 kg. How much of an effect will this additional load have on the accuracy of the closed-loop system's position tracking? [3 marks]

d) You notice that there is a leaderboard at the barcade, and, for extra crush-impressing points, you want to beat the best time to grab a thing from the claw machine. You calculate that you will need the claw to get to the required vertical position in **less than 1 second (to 2%)** to do this. It's also important to make sure that there is very little oscillation in the response, or the toy might get shaken loose from the flimsy claw. You decide to aim for **less than 5% overshoot**.

   i. Draw a rough root locus for the system, indicating the feasible region where the closed-loop poles satisfy the specifications. Hence, confirm that a complex pole pair at $-4.5 \pm 4j$ will achieve the desired results. [8 marks]

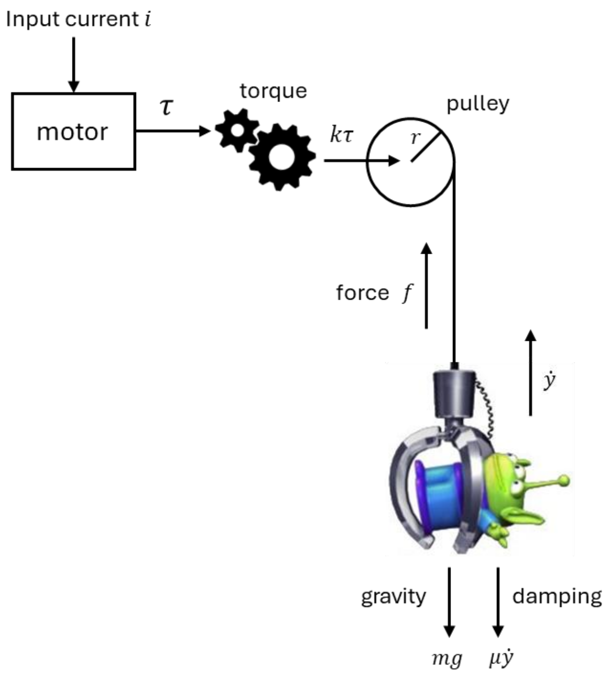
   ii. Design a lead compensator to obtain the pole pair in (d.i) in closed loop. [11 marks]

Question 3 – Information

(For convenience, you may remove this sheet by tearing or cutting on the dotted line)

Section 1 – Claw system

By disguising yourself as a maintenance worker, you are able to gain access to the inner workings of the claw machine. The schematic on the right shows the main components that move the claw up and down.



- You find the datasheet of the same electric motor online. Using the information from this, you determine that the input current i and torque τ are related by the transfer function:

$$M(s) = \frac{0.0655}{(1.2 \times 10^{-4})s + 1}$$

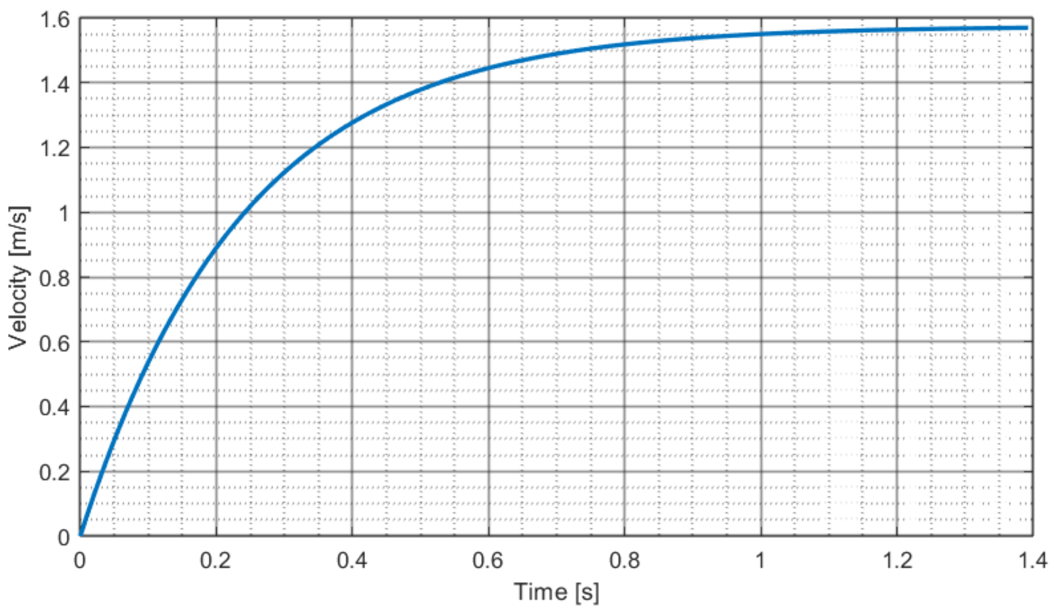
- The gearbox scales the motor's torque up by a factor of $k = 6$.

- The pulley has a radius r of 2 cm (0.02 m). The force it applies to the cable is related to the torque from the gearbox by the scalar $\frac{1}{r}$. The inertia of the pully is negligible compared to the mass of the claw.

- Besides the force from the cable, the claw is acted upon by gravity (mg), and a damping force caused by mechanical resistance in various components that is proportional and opposite to the vertical velocity $-\mu\dot{y}$. This is described by the differential equation $m\ddot{y}(t) = f(t) - \mu\dot{y}(t) - mg$. The mass of the empty claw is 1.2 kg.

Section 2 – Step Response

Starting the empty claw from rest at its lowest position ($y = 0$), you give the system an input current of 1 A. This results in the velocity response shown below. (Note that this includes the effects of gravity.)



Solution:

a) The force raising the claw is the output of the motor transfer function, scaled by the gearbox and the radius of the pulley:

$$f(s) = \frac{(0.0655)(6)\left(\frac{1}{0.02}\right)}{(1.2 \times 10^{-4})s + 1} i(s) = \frac{19.64}{(1.2 \times 10^{-4})s + 1} i(s)$$

[2 marks]

The time constant of this transfer function is much faster than that of the step response shown in the plot, so we can ignore the dynamics of the motor and replace this model with the scalar 19.64.
[2 marks]

The relationship between this force and the velocity of the claw is found by taking the inverse Laplace transform of the given differential equation, assuming initial rest:

$$\dot{y}(s) = \frac{1}{ms + \mu} = \frac{1}{1.2s + \mu} = \frac{0.833}{s + \frac{\mu}{1.2}}$$

[2 marks]

When multiplied by the scalar representing the motor, this gives

$$\frac{16.28}{s + \frac{\mu}{1.2}}$$

confirming the numerator of the given transfer function.
[2 marks]

We don't know the value of the damping parameter, but we can approximate the value of $\frac{\mu}{1.2}$ using the step response.

The final value of the response is around 1.55, so the time constant will be given by the point where it reaches (0.63)(1.55) = 0.98.
[1 mark]

This appears to be at around 0.25 seconds, giving a pole at around -4. This is close enough to confirm the time constant of the model.
[2 marks]

We must then integrate the transfer function, i.e. multiply by $\frac{1}{s}$, to give the relationship between input current and position, giving the final model. [1 mark]

b) The minimum settling time will occur when the closed-loop poles are on the asymptotes of the root locus. [1 mark]

The asymptotes are at $\frac{0-4.17}{2} = -2.08$ [1 mark], so the time constant will be $\frac{1}{2.08} = 0.48$ seconds. [1 mark]

The settling time to 2% is around $4\tau = 1.92$ seconds.

c) None. [1 mark] The system is type 1, so it will track position setpoints perfectly regardless of disturbances caused by changes to the open-loop model parameters. [2 marks]

d.i) We already worked out that the centroid of the root locus is at -2.08. There are two asymptotes so they have angles of $\pm 2\pi$. [1 mark]

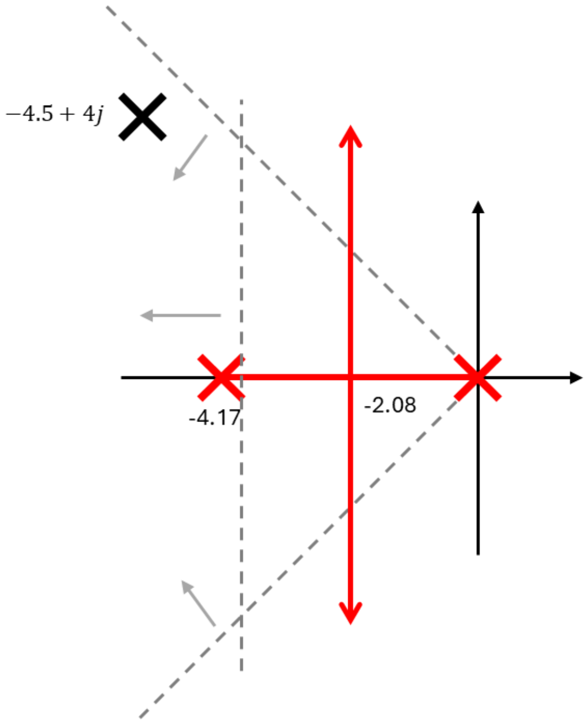
A settling time of $\tau_{2\%} \leq 1$ means $4\tau \leq 1, \therefore \tau \leq \frac{1}{4}$, so the poles will need to be left of -4. [1 mark]

Using the %OS formula, we calculate that 5% overshoot corresponds to a damping ratio of 0.7 [1 mark]. This can be represented by diagonal lines at angles of $\pm\,\text{acos}(0.7) = \pm 45.6°$ with respect to the negative real axis. [1 mark]

The suggested pole pair is within this region. [1 mark]

[1 mark for root locus, 2 marks for specification lines]

d.ii) The angle contribution from the system poles to the desired pole s_d is $-233.13°$. [2 marks]



The compensator must therefore add $\theta_c = 53.13°$ to satisfy the angle criterion at -180 degrees. [2 marks]

The zero will have an angle of θ_c on its own at $-4.5 - \frac{4}{\tan(53.13°)} = -7.5$, so the zero must be to the right of this. [1 mark]

Tracking isn't important, though, so let's choose a zero at -4.5. [1 mark for any sensible zero position.]

(Just follow the method from here for your chosen zero value.)

At -4.5, the angle from the zero is 90 degrees, [1 mark] so the pole must have an angle of 36.87 degrees. [1 mark]

This puts the pole at $-4.5 - \frac{4}{\tan(36.87°)} = -9.83$. [1 mark]

This gives the compensator $K\frac{\frac{1}{4.5}s+1}{\frac{1}{9.83}s+1}$. [1 mark]

The required gain to get a closed-loop pole at s_d is $K = \frac{1}{C(s_d)G(s_d)} = 1.12$. [1 mark]

Question 4 [30 marks]

The frequency response of a stable, third-order system is shown on the information sheet following this question.

a) What is the gain margin of this system? If it is finite, how many right-half plane poles will the closed-loop system have when this value is exceeded? Explain. [3 marks]

b) You would like the system to track position setpoints with less than 5% error. Why is it not possible to achieve this with proportional control? [2 marks]

c) Design a lag compensator to achieve the required accuracy without violating the following specifications:
   - Phase margin of $\geq 30°$
   - Sensitivity ≤ 6 dB at all frequencies
   - Sensitivity ≤ 3 dB for frequencies between 1 and 1.5 rad/s

   Explain the steps of your design, and annotate the plots as needed to support your explanations. [10 marks]

d) It may also be possible to meet the given specifications using a lead compensator. Design a lead compensator to meet the accuracy and phase margin specifications.

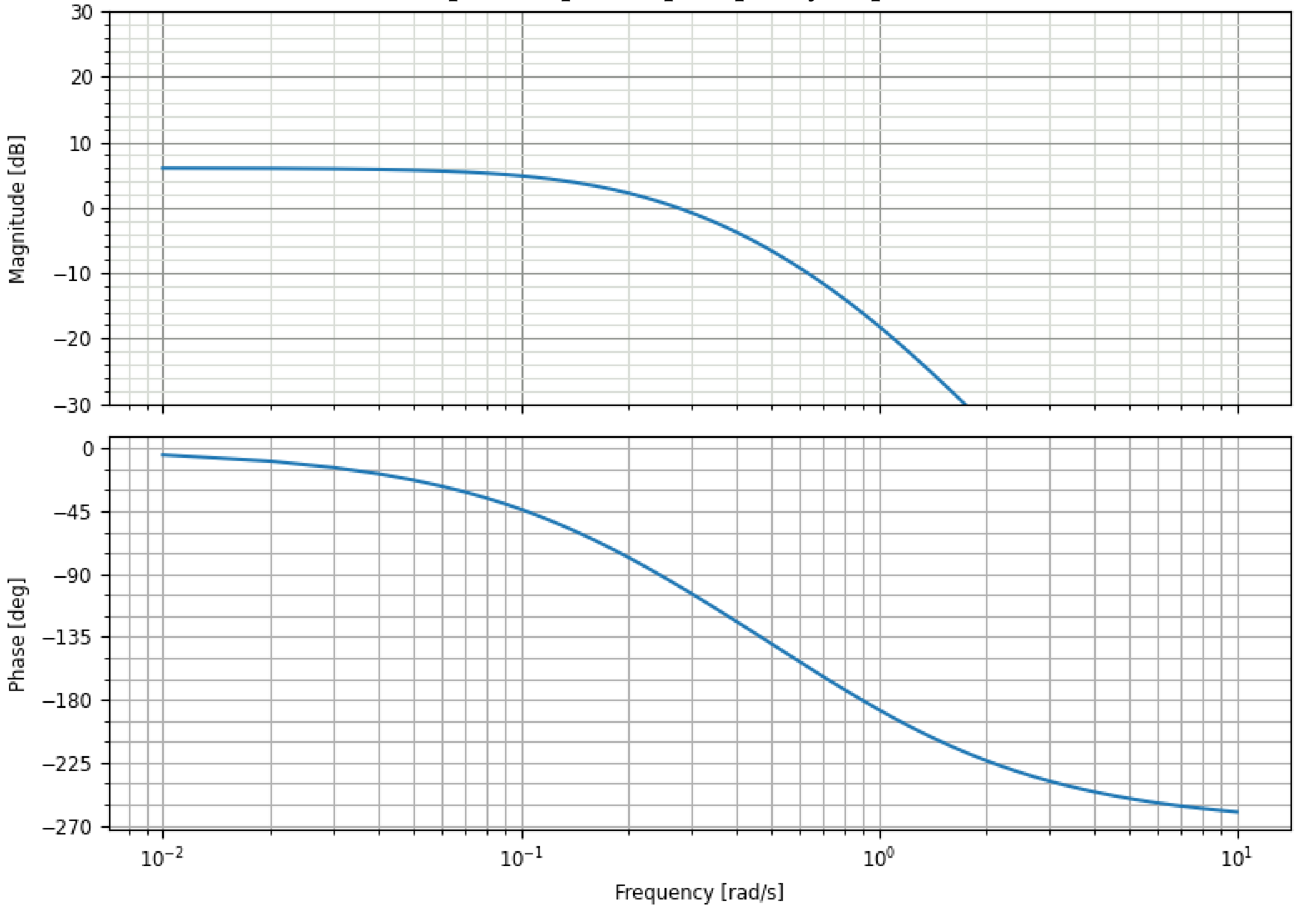
   You do not have to iterate on your design, but you should explain how you would fix the design if testing revealed that it did not achieve the specifications. [12 marks]

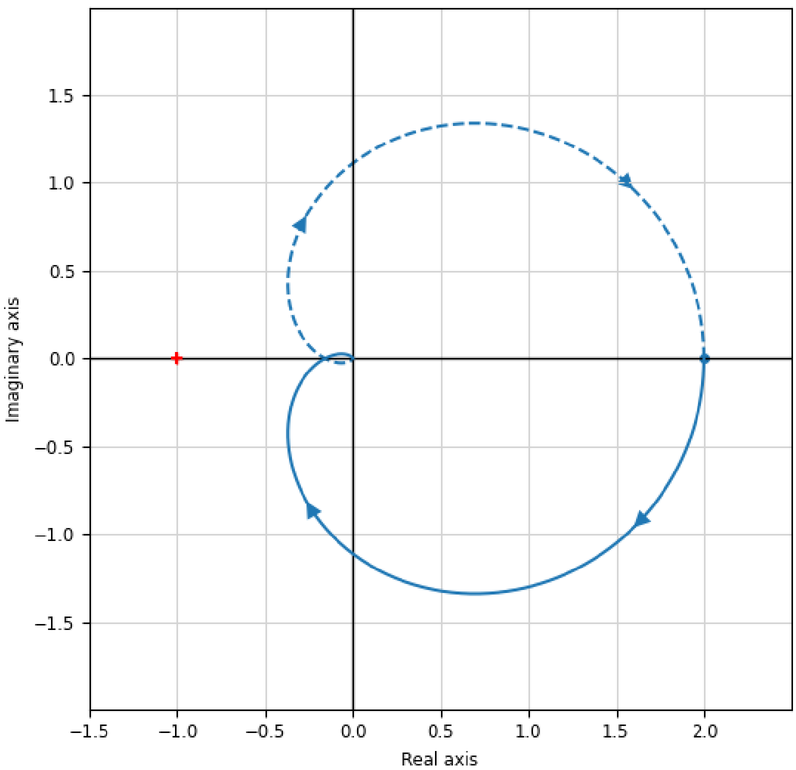
e) Which design would you recommend for an application where settling time is of critical importance? Explain. [3 marks]

Question 4 – Information

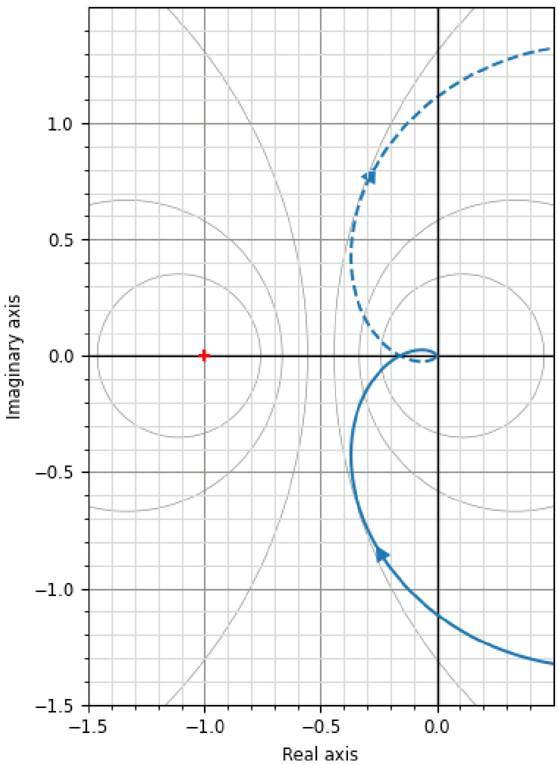
(For convenience, you may remove this sheet by tearing or cutting on the dotted line)

Bode plot of open-loop frequency response.

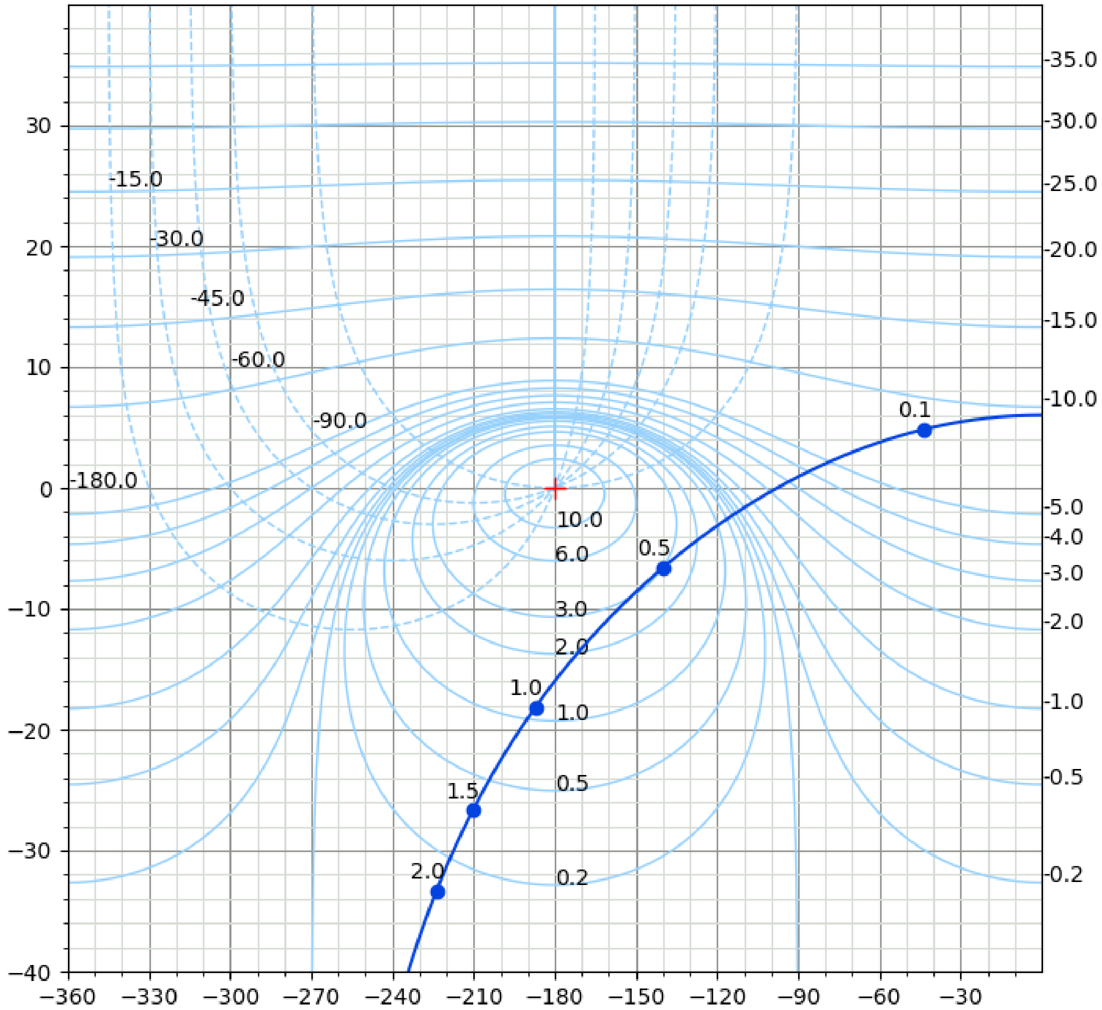




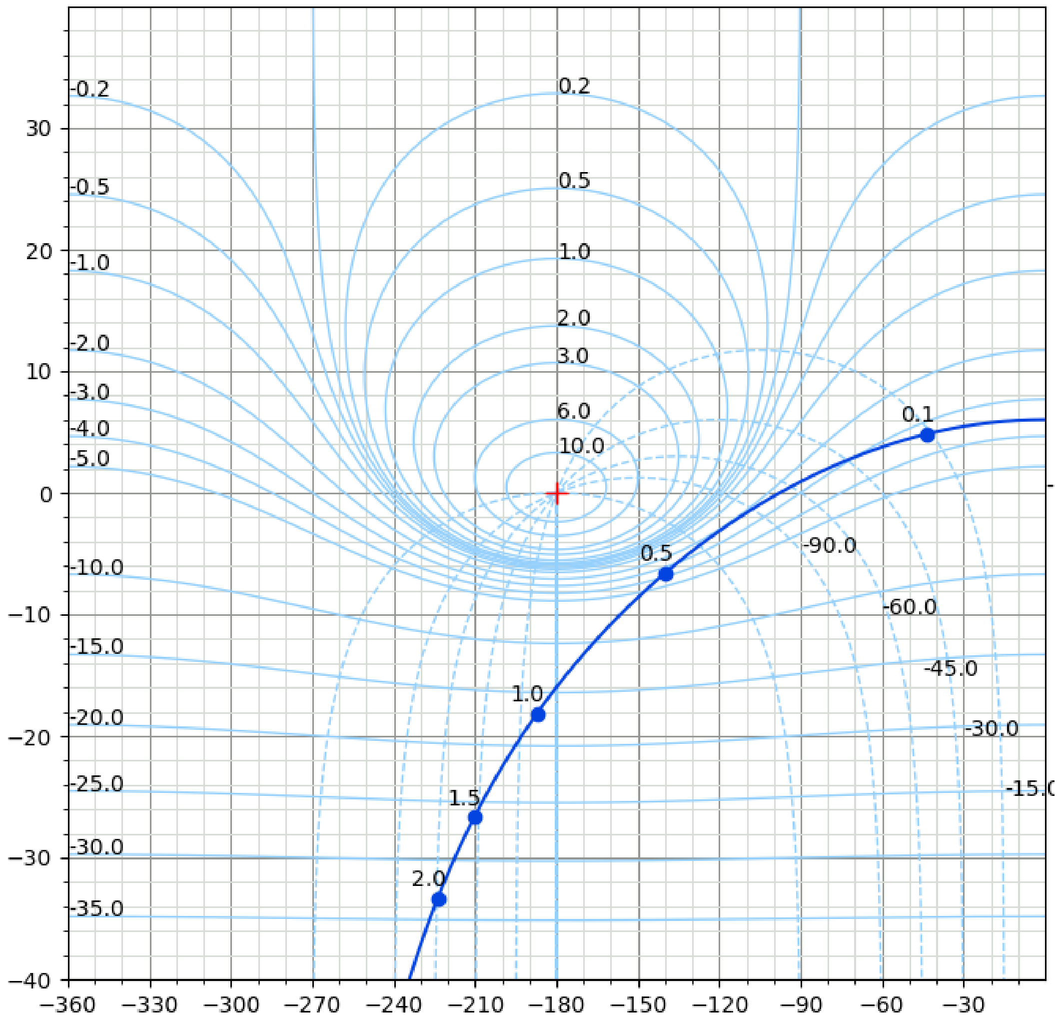
Nyquist plot



Nyquist plot - detail



Inverse Nichols chart



Nichols chart

Solutions

a) The Nyquist plot will encircle -1 twice in a clockwise direction.
From the Nyquist stability criterion, $N = P_c - P_o \therefore 2 = P_c - 0 \therefore P_c = 2$, so the system will have two unstable closed-loop poles. [2 marks]

From the Nichols chart, it is clearer to see that the gain must be increased by 16 dB (6.3) before the plot crosses -1, so this is the gain margin. [1 mark]

b) From the steady-state error $\lim_{s\to 0} s\frac{1}{s}\frac{1}{1+KG(s)} = \frac{1}{1+KG(0)} \leq 0.05$, we find that $KG(0) \geq 19$ (26 dB). This exceeds the gain margin so the closed-loop system will become unstable if this gain is applied.

c) **Gain:**
The open-loop DC gain of the system is 6 dB, so $K \geq 20$ dB (10) to meet the accuracy requirement. [1 mark]
The response with the additional gain is sketched on the inverse Nichols chart and Bode plot. [1 mark]

Required α
To find the zero/pole ratio α, we need to determine how much we need to push the high frequencies down by to avoid violating the specifications. The specification boundaries are marked on the inverse Nichols chart. [1 mark]

The marker at 1 rad/s must be pushed down by around 14 dB to clear the 3 dB circle.
Pushing the region close to the 0.5 rad/s marker down by around 12 dB will also result in the required phase margin.

We will therefore try to push frequencies above -0.5 (at the highest) down by 14 dB (5), $\therefore \alpha = 5$. [3 marks]

(This is an arbitrary design choice, follow the method from here with the value you selected.)

Pole and zero placement
We aim to place the zero at a frequency 10-15 times lower than the system's poles. [1 mark]

The corner frequency of the system's response on the Bode plot is around 0.1 rad/s, so we will try placing the zero at -0.01, and pole 5 times lower at -0.002. [2 marks]

This gives the compensator, $C(s) = 10\frac{100s+1}{500s+1}$ [1 mark]

d) We have already determined that the gain must be 10 [1 mark]

Required phase lead:
With the lead compensator, we need to determine how much to push a region of the response to the right on the inverse Nichols chart to satisfy the specifications.

We need to add about 40 degrees at the gain crossover frequency to meet the phase margin specification. [1 mark]

Using $\sin(40°) = \frac{\alpha-1}{\alpha+1}$, we find $\alpha \geq 4.6$ [1 mark]

We will increase this to $\alpha = 6$ as the additional gain added by the compensator will move the response further left at the gain crossover. [2 marks for any sensible explanation and α]

Midpoint frequency
The additional gain added at the midpoint will be $10\log_{10}(6) = 7.8$ dB [1 mark]

So, we will look for the frequency where the phase is -7.8 dB as this will become the new gain crossover.

From the bode plot, this is at around 1.6 rad/s. [2 marks for any sensible explanation and ω_m).

Using the formula $\omega_m = \frac{1}{\tau\sqrt{\alpha}}$, we find that $\tau = 0.26$. [1 mark]

This gives the compensator $C(s) = 10\frac{1.53s+1}{0.26s+1}$ [1 mark]

Recommendations
If the compensator does not add enough lead to satisfy the phase specification, the value of α can be increased. If α is already at its maximum recommended value (10) for physical implementation, the amount of phase lead could be increased by cascading two compensators. [2 marks for any suitable explanation]

e) The lead compensator. [1 mark] The lag adds a very slow pole that will then dominate the response, making the system settle much more slowly. [2 marks]